

SCI-FLT-MATCHED v0.1

Optimal matched-template map filtering

Owner-assigned package title; not a realization-level optimality claim

Scientific Rationale and Contract

STAGE B DRAFT – ALL AO-001 THROUGH AO-006 OPTIONS UNSELECTED.

No numerical route, implementation, conformity result, validation result, achieved performance, readiness, production state, scientific freeze, or Unity status is claimed.

Package version: v0.1

Author-packet date: 2026-08-31

Scientific owner for later disposition: Grant Wilson

Shared-core draft derived exclusively from the owner-approved eight-object author packet bound by manifest
SHA-256

255c66da880fc7664a57635b28a98d874fc024490d04528f802635c0382a57c8.

STAGE B DRAFT – ALL AO-001 THROUGH AO-006 OPTIONS UNSELECTED.

No numerical route, implementation, conformity result, validation result, achieved performance, readiness, production state, scientific freeze, or Unity status is claimed.

Contents

I	Scientific rationale	1
1	The scientific question	1
2	Why the estimator is normalized	1
3	Why state is frozen	1
4	Why edges are unavailable rather than repaired	1
5	Response and uncertainty are full scientific objects	2
6	Product and governance rationale	2
II	Shared normative core	2
7	Normative notation and interpretation	3
7.1	Normative language	3
7.2	Objects, domains, and indexing	3
7.3	Quantity and unit convention	4
8	Normative scientific definitions	4
8.1	Method and estimand	4
8.2	Parent classes and application locality	4
8.3	Template-response product	5
8.4	Influence support and validity	5
8.5	Weighting and parent coefficients	5
8.6	Frozen state and generations	5
8.7	Fixed-state response	6
8.8	Product roles and atomic lifecycle	6
8.9	Lifecycle states	6
8.10	First-class FLT to FRUIT handoff	7
8.11	Authority layers	7

8.12	Producer and boundary ownership	7
9	Reference estimator, response, and uncertainty	8
9.1	Authoritative operator	8
9.2	Matching-template amplitude and unbiasedness	8
9.3	Optimality criterion	8
9.4	Conditional covariance	9
9.5	Approximation observables	9
9.6	Uncertainty budget and non-combinability	9
10	Assumptions and validity domain	10
10.1	Validity table	11
11	Normative requirements	12
12	Falsifiable scientific predictions	15
13	Unselected authored option sets	17
13.1	SCI-FLT-MATCHED-A0-001: Weighting and noise authority	17
13.1.1	SCI-FLT-MATCHED-A0-001-A: exact inverse-covariance GLS	17
13.1.2	SCI-FLT-MATCHED-A0-001-B: structured covariance-derived weighting	18
13.1.3	SCI-FLT-MATCHED-A0-001-C: radially symmetrized average map-noise PSD weighting	18
13.1.4	SCI-FLT-MATCHED-A0-001-D: declared weaker positive-semidefinite weighting . .	19
13.2	SCI-FLT-MATCHED-A0-002: Quantitative approximation envelope	20
13.2.1	SCI-FLT-MATCHED-A0-002-A: exact-identity envelope	20
13.2.2	SCI-FLT-MATCHED-A0-002-B: strict uniform scientific envelope	21
13.2.3	SCI-FLT-MATCHED-A0-002-C: named-use stratified envelope	21
13.3	SCI-FLT-MATCHED-A0-003: Conditional covariance representation	22
13.3.1	SCI-FLT-MATCHED-A0-003-A: exact explicit covariance	22
13.3.2	SCI-FLT-MATCHED-A0-003-B: exact structured covariance	23
13.3.3	SCI-FLT-MATCHED-A0-003-C: authorized projected covariance	23
13.3.4	SCI-FLT-MATCHED-A0-003-D: exact lineage-resolvable covariance on demand . .	23
13.3.5	SCI-FLT-MATCHED-A0-003-E: covariance unavailable	24
13.4	SCI-FLT-MATCHED-A0-004: Immutable state persistence	24
13.4.1	SCI-FLT-MATCHED-A0-004-A: full state materialization	24
13.4.2	SCI-FLT-MATCHED-A0-004-B: structured compact state	25
13.4.3	SCI-FLT-MATCHED-A0-004-C: exact lineage reconstruction	25

13.5	SCI-FLT-MATCHED-A0-005: Response, product, and FLT to FRUIT interface	26
13.5.1	SCI-FLT-MATCHED-A0-005-A: full response materialization	26
13.5.2	SCI-FLT-MATCHED-A0-005-B: exact structured response with documented query contract	26
13.5.3	SCI-FLT-MATCHED-A0-005-C: exact lineage-resolvable response on demand	27
13.6	SCI-FLT-MATCHED-A0-006: VAL profile granularity	27
13.6.1	SCI-FLT-MATCHED-A0-006-A: six role-separated profiles	27
13.6.2	SCI-FLT-MATCHED-A0-006-B: three layered profiles with mandatory subverdicts .	28
13.6.3	SCI-FLT-MATCHED-A0-006-C: one composite profile with a six-role verdict vector	28
13.7	Option-set closure and unavailable routes	29
14	Edge cases, exclusions, and failure routing	29
14.1	Explicit method exclusions	31
14.2	Nonclaim closure	32
III	Owner-disposition guide	32
15	What remains for the scientific owner	32
A	Author-packet basis and firewall statement	33

Part I

Scientific rationale

1 The scientific question

The method answers one narrow question: at each admitted map location, what is the amplitude of one exact, already-declared template in one exact ordinary-MAP parent? The answer is a map of template amplitude. The method does not decide whether a source exists, choose a candidate, infer a posterior sky, or attach catalog meaning to a peak.

The template is not merely a shape used for smoothing. It is the expected parent-map response per unit amplitude, including its scale, phase, anchor, sampling, support, and calibration/beam lineage. That scaling defines what one unit of the answer means. A template whose unit amplitude denotes point-source flux can support that interpretation only when the exact CAL/BEAM and parent lineages authorize it; another template gives a shape amplitude instead.

2 Why the estimator is normalized

The parent is contracted with the template through a frozen weighting object. The same weighting is applied to the template itself, producing the normalization. Dividing these two contractions makes an exact matching template return its supplied amplitude on complete valid support. This unit response is the central scientific identity.

Weighting and uncertainty are deliberately not conflated. If the weighting is the exact authorized inverse covariance on the same population, support, conditioning state, and estimable subspace, the estimator has the familiar inverse-covariance GLS minimum-variance property. Under a weaker spectral or reliability weight, it can remain matching-template normalized without being minimum variance. In that case the denominator is normalization, not a variance estimate.

3 Why state is frozen

The transformation must mean the same thing for the signal parent and every compatible NOI realization member. State is therefore declared externally or learned once from the exact parent, assigned an immutable generation, and frozen. Learning again from each NOI member would change the ensemble method and its uncertainty meaning. New NOI-informed knowledge creates a successor generation rather than rewriting an existing one.

Observation and coadd parents also remain scientifically separate. Filtering an observation and filtering a coadd are two local applications with different parent identities and state generations. This contract neither asserts that filtering commutes with coaddition nor owns a coaddition operation.

4 Why edges are unavailable rather than repaired

Each amplitude depends on a complete influence support: every parent sample used by the template, weighting, window, boundary action, and normalization. If one dependency is absent or invalid, the estimator defined here does not exist at that location. Filling, extending, renormalizing, fitting a

background, or learning an edge taper would define another method. Numerical fill is harmless only after erosion proves that no admitted output depends on it.

5 Response and uncertainty are full scientific objects

Unit self-response does not make the response a single symmetric beam. The off-diagonal response can be anisotropic, asymmetric, nonstationary, position-dependent, or nonlocal. Reducing it to a kernel, radial profile, or solid angle is allowed only after the required invariances, measure, and valid domain are established.

Likewise, a conditional covariance product exists only when matching parent covariance authority exists. Calibration, learned-state, approximation, representation, NOI sampling, and template-mismatch contributions have different populations and conditioning. The normative uncertainty budget keeps them separate rather than producing an unsupported scalar total.

6 Product and governance rationale

The primary science product is one atomic signal bundle: amplitude map plus the facts required to know exactly what it means. Covariance, empirical NOI, response, and state/lineage products are independently atomic companions. Their failure is explicit. Optional failure does not silently alter the signal estimand, while failure of a required signal member prevents publication of a complete signal bundle.

The product also preserves a first-class FLT to FRUIT handoff. It makes every FLT-side scientific fact available or exactly reconstructable without exposing undocumented implementation state. It intentionally says nothing about FRUIT's future model, iteration, selection, response, uncertainty, or interpretation.

Finally, the contract separates six layers that are often confused: scientific authority defines the method; conformity concerns a specific implementation; validation evaluates immutable profiles; achieved performance is measured behavior; readiness is a governance decision; and production is an operational state. Evidence in one layer does not promote another.

Part II

Shared normative core

The following six modules are imported unchanged into both the scientist- and engineering-facing documents. They are the single normative source for all definitions, equations, assumptions, requirements, predictions, option alternatives, failure rules, and validation consequences.

7 Normative notation and interpretation

7.1 Normative language

“Shall” is a contract requirement. “May” identifies an allowed form whose scientific preconditions still have to be met. “Exact” means exact with respect to the declared mathematical realization and its declared numerical identity; it does not mean that nature, calibration, or the parent data are known without uncertainty. “Unknown” and “unavailable” never mean zero. An option in Section 13 is a decision candidate, not a selected route.

7.2 Objects, domains, and indexing

An application location is denoted by x and a parent-map sample by z . Both are coordinates in one declared map-domain chart with an immutable WCS, frame, grouping, pixel measure, and indexing order. The admitted parent field is real-valued, $m : \mathcal{D}_m \rightarrow \mathbb{R}$, and the unit-amplitude template centered or anchored at x is real-valued, $t_x : \mathcal{S}_t(x) \rightarrow \mathbb{R}$. The physical template amplitude A , the published amplitude $\hat{A}$, and the contractions N and D are real scalars. Let $\mathcal{X}_{\text{in}}$ be the real vector space of admitted parent samples, $\mathcal{X}_{\text{out}}$ the real vector space of published amplitude samples, and $\mathcal{X}_{\text{anc}}$ the real vector space indexing template anchors. Then $L : \mathcal{X}_{\text{in}} \rightarrow \mathcal{X}_{\text{out}}$ and $R_t : \mathcal{X}_{\text{anc}} \rightarrow \mathcal{X}_{\text{out}}$. The restriction of any field u to the samples required at x is u_x . The realized linear weighting operator is a real linear map Q_x on the admitted parent-sample space, and its declared null space is $\mathcal{N}(Q_x)$.

The bracket

$$\langle a, b \rangle_x = \sum_{z \in \mathcal{S}_t(x)} a(z)^* b(z) \mu(z)$$

denotes the declared discrete inner product. A continuum representation may replace the sum by an integral only when the measure, sampling relation, and equivalence to the published discrete estimand are declared. Here $\mu(z)$ is the declared pixel or quadrature measure. Complex conjugation is explicit so the same notation can be used for a complex transform representation. Such a representation may use coefficients in $\mathbb{C}$, but it shall preserve the conjugate symmetry of real fields, use the Hermitian adjoint for transform- domain operators, and induce the same real Q_x , N , D , L , and $\hat{A}$ after inverse transformation. Complex transform coefficients do not change the real scientific codomains above. In the real sample basis, covariances are real symmetric matrices and ordinary transpose is therefore the covariance adjoint. The contract does not assume stationarity, isotropy, periodicity, a flat sky, or a Euclidean pixel metric unless the selected realization declares and validates it.

Symbol	Normative meaning
m, m_x	Immutable admitted ordinary-MAP observation or coadd parent, and its complete restriction needed at x .
t_x	Immutable materialized parent-map response for unit template amplitude at x ; its scaling fixes the amplitude convention.
Q_x	Frozen realized weighting operator. It is inverse covariance only when exact covariance authority establishes that identity.
$N(x)$	Weighted template-parent contraction $\langle t_x, Q_x m_x \rangle_x$.
$D(x)$	Weighted template self-contraction $\langle t_x, Q_x t_x \rangle_x$; normalization unless exact inverse-covariance GLS premises establish a variance interpretation.
$\hat{A}(x)$	Published template-amplitude estimand $N(x)/D(x)$ at an admitted location.

Symbol	Normative meaning
L_x	Fixed-state linear functional that maps a compatible parent field to $\hat{A}(x)$.
$R_t(x, y)$	Fixed-state response $L_x t_y$ to a unit template anchored at y .
C_{parent}	Authorized real symmetric parent covariance on $\mathcal{X}_{\text{in}}$ for an explicitly named population and conditioning state.
C_{cond}	Conditional covariance of the filtered amplitude field, $LC_{\text{parent}}L^T$ on $\mathcal{X}_{\text{out}}$, when every premise is authorized.
g	Immutable method/state generation. A successor is a distinct identity.
$\mathcal{V}$	Set of locations satisfying every parent, template, support, operator, normalization, unit, calibration, and selected-option predicate.

7.3 Quantity and unit convention

The template is the expected parent-map response per unit amplitude. Therefore

$$[\hat{A}] = [m]/[t].$$

A point-source flux interpretation exists only if the template producer and CAL/BEAM lineage establish that exact amplitude convention. Otherwise $\hat{A}$ is a shape amplitude. Parent beam, DC or integral response, extended-source fidelity, matched-filter solid angle, and calibration covariance are not inherited by implication.

8 Normative scientific definitions

8.1 Method and estimand

SCI-FLT-MATCHED is an optimal matched-template amplitude estimator only under the exact criterion and premises in Section 9.3. It estimates the local amplitude of one exact declared template in one immutable admitted ordinary-MAP parent. It publishes a filtered amplitude map. It does not detect, select, fit, deblend, catalog, classify, or interpret sources, and it is neither posterior/Wiener sky reconstruction nor ordinary convolution.

Each realization has the immutable scientific identity

```
(method_id, estimand, parent_id, parent_grouping, domain, WCS/frame, template_id,
state_generation, weighting_object, operator_realization, approximation, response,
support/null, units/calibration, covariance, NOI_lifecycle, product_bundle,
validity, failure_policy, provenance).
```

Any scientifically consequential change to one of these fields creates a different realization, generation, or method as classified by the selected SCI-FLT-MATCHED-A0-002 and SCI-FLT-MATCHED-A0-004 dispositions.

8.2 Parent classes and application locality

Exactly one of two parent classes is admitted per application:

1. one immutable normalized ordinary-MAP observation bundle; or

2. one immutable normalized ordinary-MAP coadd bundle with its exact contributing-observation membership.

They are distinct local applications. This contract asserts no equality, commutation, or interchangeability between “filter then coadd” and “coadd then filter”, and owns no coaddition. JINC and other derived parents are not admitted. Numerical computability cannot repair a parent that lacks required MAP authority.

8.3 Template-response product

Each application has exactly one immutable template-response product. It is scientifically declared outside target, source, candidate, population, and NOI-member learning. The product binds its materialized samples, amplitude scaling, anchor and phase convention, parent-map response quantity, WCS/frame, sampling, support, CAL/BEAM lineage, validity, and generation. Template mismatch is not silently absorbed into amplitude uncertainty or estimator normalization.

8.4 Influence support and validity

The influence support at x is the complete set of parent samples on which $N(x)$, $D(x)$, the realized Q_x , the template, any window, and any declared nonlocal operation depend. A location is admitted only if that complete set is available and valid. No partial-support estimator, edge renormalization, extension, imputation, background subtraction, learned taper, or signal-derived support belongs to this contract. Numerical fill values may be used internally only after exact erosion proves that no admitted output depends on them. Affected locations are **unavailable**, not zero.

8.5 Weighting and parent coefficients

Q_x is a declared frozen weighting object chosen from a later owner-disposed **SCI-FLT-MATCHED-A0-001** alternative. A parent coefficient has only its exact declared role in that alternative. Its name, positivity, units, reciprocal, or diagonal placement does not make it covariance, precision, independence, or inverse variance. Rank, null space, regularization, support, edge behavior, state population, and units are part of Q_x identity.

8.6 Frozen state and generations

Requested, effective, observation-resolved, realized, and published state are distinct objects and shall retain distinct identities and provenance even when some fields happen to be numerically equal.

For generation g ,

$$\text{state}_g = \text{DeclareOrLearnOnce}(\text{parent}_g, \text{external inputs}_g), \quad \text{product}_g = \text{Apply}(\text{parent}_g; \text{state}_g).$$

State is externally declared or learned once from the exact parent and then frozen. Observation and coadd states are separate. Every compatible NOI realization member receives exactly the same frozen transformation, support, normalization, and failure rules. NOI-informed updating produces an immutable successor generation. Per-member relearning is a different future method and is unavailable here.

8.7 Fixed-state response

The response is the complete fixed-state map $R_t(x, y) = L_x t_y$. It may be asymmetric, anisotropic, nonstationary, position-dependent, or nonlocal. A single response kernel, radial profile, beam, or solid angle is scientifically sufficient only when every invariance and measure needed for that reduction has been established. Matching-template unit response at admitted positions does not establish fidelity to other morphologies.

8.8 Product roles and atomic lifecycle

The required atomic signal bundle contains, as one role-complete unit:

1. the filtered amplitude field and its quantity and units;
2. exact parent identity, class, generation, grouping, WCS/frame, coadd membership when applicable, support, and parent provenance;
3. template identity, amplitude convention, response reference, and CAL/BEAM lineage;
4. method, state generation, Q_x , operator realization, approximation identity, support/null rules, and validity mask;
5. uncertainty availability and the identity of any qualified companion; and
6. lifecycle state, failure semantics, producer identity, and complete lineage needed to distinguish or reconstruct the realization.

Failure of any member fails the whole signal bundle.

Independently atomic qualified companions may represent authorized C_{cond} , a frozen-NOI uncertainty estimand, response materialization, or exact state/lineage required by a named use. Each companion has its own estimand, population, unit, support, validity, lifecycle, and policy. Its absence or failure is explicit and does not alter the signal estimand or retroactively invalidate a complete signal bundle unless an exact named-use policy requires it.

No numerator, denominator, kernel, PSD, response slice, inverse scale, standardized field, or intermediate is a public science product without an exact role, meaning, unit, validity, lifecycle, and policy.

8.9 Lifecycle states

State	Meaning
disabled	The method or qualified product was explicitly not requested; there is no attempted estimand.
unavailable	A required scientific premise, authority, input, selected option, or valid support is absent; no numerical value is asserted.
failed	An authorized attempt began but a required operation, closure check, or atomic member failed.
complete	Every required member and predicate for the exact role and generation is present and passed; this is not a performance or readiness claim.
superseded	An immutable successor exists. The predecessor remains identifiable and is not silently rewritten.

8.10 First-class FLT to FRUIT handoff

The product lineage shall let a future separately authorized FRUIT method consume the filtered amplitude product without reconstructing undocumented FLT implementation state. At minimum it identifies the filtered quantity, template, full response reference, valid support, units/calibration, method/state identity, uncertainty availability, and provenance. Additional authorized FLT state shall be exactly available or reconstructable through lineage. This handoff defines no FRUIT source model, subtraction/add-back, recurrence, learning, stopping, restart, selection, response, uncertainty, or interpretation.

8.11 Authority layers

This document is scientific authority for definitions and requirements after owner disposition and freeze. An implementation may later claim conformity only through auditable evidence against a selected contract. Validation may evaluate immutable profiles but does not invent producer facts or policy. Measured behavior is achieved performance, not scientific authority. Readiness and production are separate governance decisions. No draft, option, test result, or lifecycle state collapses these layers.

8.12 Producer and boundary ownership

Owner		Normative boundary
FLT	/	Owns the template-amplitude transformation, declared template use, learn-once state definition and realization, resulting response and support consequences, product roles, lifecycle, failure semantics, named-use policies, fixed transformation supplied to compatible NOI members, and preservation of the FLT to FRUIT handoff.
SCI-FLT-MATCHED		
MAP	pro-	Owns the immutable admitted parent map bundle, generation, grouping, WCS/frame, support/validity, signal quantity and unit, response/calibration provenance, and exact contributing-observation membership of a coadd. FLT may consume but not rewrite these facts.
Template		Owns the exact immutable materialized template-response product, including scale, phase, anchor, sampling, support, lineage, and validity.
CAL/BEAM		Own calibrated transfer, amplitude-convention inputs, beam lineage, and calibration covariance. FLT owns only their declared use and the resulting template-amplitude response.
NOI		Owns realization generation and empirical-uncertainty meanings. FLT owns the identical frozen transformation applied to compatible members.
VAL		May register and evaluate exact immutable profiles. FLT owns the producer facts, named-use policies, required actions, and failure consequences; VAL may not invent them.
FRUIT		Owns any future source model, subtraction/add-back, recurrence, learning, stopping/restart/selection, response, uncertainty, and interpretation. FLT owns only preservation of the first-class scientific handoff.
SRC		No present ownership boundary or dependency is created; source analysis is outside scope.

9 Reference estimator, response, and uncertainty

9.1 Authoritative operator

For every admitted $x \in \mathcal{V}$, the scientific reference is

$$N(x) = \langle t_x, Q_x m_x \rangle_x, \quad (1)$$

$$D(x) = \langle t_x, Q_x t_x \rangle_x, \quad (2)$$

$$\hat{A}(x) = \frac{N(x)}{D(x)}, \quad (3)$$

$$L_x u = \frac{\langle t_x, Q_x u_x \rangle_x}{D(x)}, \quad (4)$$

$$R_t(x, y) = L_x t_y. \quad (5)$$

$D(x)$ shall be finite, real to the declared numerical tolerance, and strictly positive under the selected realization. Otherwise x is **unavailable** or the required bundle is **failed** according to whether the defect precedes or occurs during an authorized attempt. Neither outcome is amplitude zero.

9.2 Matching-template amplitude and unbiasedness

Let the admitted parent near y be

$$m = A t_y + n,$$

where A has the declared template-amplitude unit and the frozen-state noise obeys $E[n \mid \text{state}_g] = 0$ on complete influence support. Linearity gives

$$E[\hat{A}(x) \mid \text{state}_g] = A R_t(x, y).$$

At $x = y$ the exact reference operator satisfies

$$R_t(y, y) = \frac{\langle t_y, Q_y t_y \rangle_y}{D(y)} = 1, \quad E[\hat{A}(y) \mid \text{state}_g] = A.$$

This is conditional fixed-state amplitude unbiasedness for a matching supplied template. It does not assert unbiasedness after target-dependent state learning, under template mismatch, for incomplete support, or for a different sky estimand.

9.3 Optimality criterion

“Optimal matched-template map filtering” is the exact owner-assigned package title. Citation of that proper name is not a realization-level optimality claim and shall be accompanied by the draft/unselected status until owner disposition. Outside that proper-name citation, the unqualified word “optimal” is forbidden before the following criterion is authorized for the exact realization.

The allowed exact claim is: among fixed-state linear estimators of the declared template amplitude that are unbiased for t_x , L_x has minimum conditional variance when C_{parent} is the exact authorized covariance on the admitted support, Q_x is its authorized inverse on the estimable subspace, the null-space and constraints are handled identically, and all domain, population, conditioning, unit, and regularization premises hold. This is the inverse-covariance generalized least-squares (GLS) criterion. Under weaker weighting, the method remains normalized and matching-template amplitude-unbiased but no minimum-variance or covariance-authority claim follows.

9.4 Conditional covariance

When an exact matching parent covariance is authorized, the fixed-state conditional covariance is

$$C_{\text{cond}}(x, y) = \text{Cov}[\hat{A}(x), \hat{A}(y) \mid \text{state}_g] = L_x C_{\text{parent}} L_y^T. \quad (6)$$

Only under the exact inverse-covariance GLS premises above may $D(x)^{-1}$ be identified with the marginal conditional variance at x . Otherwise D is normalization. A diagonal representation of C_{cond} does not assert independence; omitted entries remain unknown.

9.5 Approximation observables

Let $\tilde{L}$, $\tilde{D}$, $\tilde{R}_t$, and $\tilde{C}_{\text{cond}}$ denote an approximate realization compared with the exact reference on a declared test domain $\mathcal{T} \subseteq \mathcal{V}$. The minimum common observables are

$$e_D(x) := \frac{|\tilde{D}(x) - D(x)|}{D(x)}, \quad (7)$$

$$e_A(x) := |\tilde{R}_t(x, x) - 1|, \quad (8)$$

$$e_R(x, y) := |\tilde{R}_t(x, y) - R_t(x, y)|, \quad (9)$$

$$\Delta_{\text{supp}} := \mathbf{1}\{\tilde{\mathcal{V}} \neq \mathcal{V}\}, \quad (10)$$

$$\Delta_{\text{null}} := \mathbf{1}\{\mathcal{N}(\tilde{L}) \neq \mathcal{N}(L)\}, \quad (11)$$

$$e_C := \frac{\|P_{\text{out}}(\tilde{C}_{\text{cond}} - C_{\text{cond}})P_{\text{out}}^T\|}{\|P_{\text{out}}C_{\text{cond}}P_{\text{out}}^T\|}, \quad (12)$$

where P_{in} , P_{out} , and P_{anc} are separately declared projections (or identity maps) on $\mathcal{X}_{\text{in}}$, $\mathcal{X}_{\text{out}}$, and $\mathcal{X}_{\text{anc}}$, respectively. Every norm, domain measure or grid, aggregation, unresolved-mode set, projection codomain, and zero-denominator rule is declared by the selected **SCI-FLT-MATCHED-A0-002** option. Support and null-space equality are scientific predicates, not merely small numeric residuals.

9.6 Uncertainty budget and non-combinability

The uncertainty budget is role-separated:

ID	Contribution	Contract treatment
U1	Parent stochastic covariance	Propagate as $LC_{\text{parent}}L^T$ only for the exact authorized population and conditioning state.
U2	Calibration and template-amplitude scale	Preserve joint CAL/BEAM and parent-template covariance authority. Missing cross-covariance is unavailable , not zero.
U3	Learned-state uncertainty	Base products condition on the frozen state. Marginalization over state learning requires separate authority and cannot be inferred from U1.
U4	Operator approximation discrepancy	Bound only through a selected SCI-FLT-MATCHED-A0-002 envelope. An unbounded approximation makes affected uncertainty use unavailable .

ID	Contribution	Contract treatment
U5	Covariance representation or projection	Declare lost modes, omitted correlations, reconstruction fidelity, and consumer limits through SCI-FLT-MATCHED-AO-003.
U6	Frozen-NOI empirical uncertainty	A qualified companion with its own population, finite-sample estimator, dependence, and validity. It is not automatically C_{parent} , C_{cond} , significance, or physical noise.
U7	Template mismatch and unmodeled systematics	Outside the base matching-template covariance claim. A named extension requires separate scientific authority; otherwise affected interpretation is unavailable.

No scalar “total uncertainty” may combine these rows unless every term, cross-term, conditioning relation, unit, and population is authorized. The absence of a term is disclosed as unavailable, never silently set to zero.

10 Assumptions and validity domain

The unbiasedness, response, optimality, and covariance statements are valid only on the intersection of the following declared predicates. Every product shall identify which predicates were evaluated, by whose authority, and with what immutable evidence.

ID	Assumption or predicate
ASM-001	The parent is exactly one immutable normalized ordinary-MAP observation or coadd bundle with authoritative identity, generation, grouping, WCS/frame, support, units, calibration provenance, and coadd membership when applicable.
ASM-002	Parent, template, physical amplitude, filtered amplitude, and sample-basis covariance have the declared real codomains; the chart, indexing order, sampling, pixel measure, and coordinate transformations are common to parent, template, operator, and response, or an exact declared mapping relates them. Any complex transform representation preserves real- field conjugacy and uses the corresponding Hermitian adjoint.
ASM-003	t_x is the one immutable template-response product for the application and expresses expected parent-map response per unit declared amplitude, with fixed scale, anchor, and phase.
ASM-004	The complete influence support is present and valid. No admitted output depends on fill, extension, imputation, partial support, edge renormalization, adaptive background, learned taper, or signal-derived support.
ASM-005	Q_x is the exact frozen realized weighting object named by the selected SCI-FLT-MATCHED-AO-001 alternative, including domain, population, units, rank, null space, regularization, window, and edge rules.
ASM-006	$D(x)$ is finite, real within the declared numerical identity tolerance, and strictly positive.

ID	Assumption or predicate
ASM-007	For amplitude unbiasedness, the admitted noise has zero conditional mean on the complete support given the frozen state and does not induce target-dependent changes in the template or operator.
ASM-008	For the GLS optimality claim, C_{parent} is the exact authorized covariance for the population, support, conditioning, and constraints, and Q_x is its exact inverse on the same estimable subspace.
ASM-009	For C_{cond} , L , C_{parent} , response, support, units, and conditioning state refer to the same realization and generation.
ASM-010	The state is externally declared or learned once from the exact parent and external inputs, then frozen. It is neither target-updated nor NOI-member releared.
ASM-011	A compatible NOI ensemble member receives the identical state, operator, support, normalization, approximation, and failure policy as the signal parent.
ASM-012	Any approximate operator satisfies a selected and fully parameterized SCI-FLT-MATCHED-A0-002 envelope on its complete declared test domain.
ASM-013	Response and uncertainty representations satisfy selected SCI-FLT-MATCHED-A0-003 and SCI-FLT-MATCHED-A0-005 reconstruction, omitted-mode, consumer, and lifecycle limits.
ASM-014	Template and parent calibration lineages are compatible; any claimed point-source flux meaning or calibration covariance has explicit CAL/BEAM authority.
ASM-015	A method option, named-use policy, and lifecycle route are explicitly selected; there is no automatic selector, fallback, or silent substitution.

10.1 Validity table

Dimension	Admission evidence	Invalid or missing case	Required outcome
Parent identity	Exact bundle identity, class, generation, grouping, WCS/frame, support, unit, quantity, calibration lineage, and coadd membership	Missing, mutable, derived, or mixed parent	unavailable before application; no FLT repair or coaddition.
Application class	Observation or coadd declared independently	Assumed commutation or borrowed state between classes	unavailable ; require a distinct realization.
Template	Exact materialized response per unit amplitude with scale, phase, anchor, sampling, and lineage	Learned from target/member, mismatched chart, ambiguous scale, or missing lineage	unavailable ; no inferred template.
Influence support	Complete dependency closure and erosion proof	Missing, invalid, filled, extended, imputed, or partially supported dependency	Mark affected locations unavailable ; no partial estimate.

Dimension	Admission evidence	Invalid or missing case	Required outcome
Weighting state	Selected SCI-FLT-MATCHED-A0-001, population, operator, rank/null, regularization, coefficients, window and edge rules bound to g	Unknown, unselected, incompatible, or changing Q_x	Fail closed; no fall-back.
Normalization	Finite, real-to-tolerance, strictly positive $D(x)$	NaN, infinite, complex beyond tolerance, zero, negative, or null-confounded	Location unavailable or attempted bundle failed ; never write zero.
Approximation	Selected SCI-FLT-MATCHED-A0-002 and all metrics within bounds on the full test domain	Missing exact comparator, uncovered modes, support/null mismatch, or any exceeded bound	failed for that realization; material change is a new realization or method.
Response	Full fixed-state response or selected exact reconstruction contract	Undocumented kernel, assumed invariance, or unreconstructable response	Response-dependent product or handoff unavailable .
Units and calibration	$[\hat{A}] = [m]/[t]$ and compatible joint lineage	Unjustified flux, beam, solid angle, DC, extended-source, or covariance meaning	Forbid the claim; required named use is unavailable .
Covariance	Matching authorized C_{parent} and selected SCI-FLT-MATCHED-A0-003	Missing population/conditioning, omitted correlations treated as zero, or D^{-1} used without GLS	Covariance use unavailable or companion failed ; signal may remain complete .
NOI parity	Identical frozen transformation for compatible members	Per-member relearning, different support, or mixed generation	NOI companion unavailable or failed ; separate method required.
Lifecycle	Atomic membership, named-use policy, and immutable state	Mixed status, partial signal bundle, silent mutation, or overwritten predecessor	Bundle failed ; retain predecessor and successor identities.
FLT to FRUIT	Selected SCI-FLT-MATCHED-A0-005, documented response/state lineage, and handoff profile	Requires undocumented implementation state or supplies FRUIT science	Handoff unavailable ; signal product status is unchanged.

The valid output domain $\mathcal{V}$ is the intersection of all location-level predicates above. Bundle-level validity additionally requires all atomic identity and lifecycle members. Consumer-specific validity is never broader than producer validity and may be narrower under a named-use policy.

11 Normative requirements

Requirement	Normative statement
SCI-FLT-MATCHED-REQ-001	The method identity shall be SCI-FLT-MATCHED , titled “Optimal matched-template map filtering,” subject to the qualified optimality rule in Section 9.3.
SCI-FLT-MATCHED-REQ-002	The sole estimand shall be local amplitude of one exact supplied template; the public signal shall be a filtered amplitude map.
SCI-FLT-MATCHED-REQ-003	Each application shall bind exactly one immutable normalized ordinary-MAP observation parent or exactly one immutable normalized ordinary-MAP coadd parent. JINC and other derived parents are excluded.
SCI-FLT-MATCHED-REQ-004	Observation and coadd applications shall remain distinct; no commutation, equivalence, state sharing, or FLT-owned coaddition may be asserted.
SCI-FLT-MATCHED-REQ-005	Each application shall bind one immutable materialized template-response product expressing expected parent-map response per unit amplitude, with exact scale, phase, anchor, sampling, support, and lineage.
SCI-FLT-MATCHED-REQ-006	The template shall not be learned or updated from a target, source, candidate, population, or NOI realization member.
SCI-FLT-MATCHED-REQ-007	The realization shall declare domain, real scientific codomains, WCS/frame, indexing, measure, inner product, template scaling, phase, units, support, boundaries, rank/null space, regularization, validity, and any complex-transform conjugacy and Hermitian-adjoint convention.
SCI-FLT-MATCHED-REQ-008	The authoritative estimator shall be exactly N/D and L_x as defined in Section 9.
SCI-FLT-MATCHED-REQ-009	Every admitted $D(x)$ shall be finite, real to declared tolerance, and strictly positive; an invalid D is typed unavailability or failure, never amplitude zero.
SCI-FLT-MATCHED-REQ-010	Complete declared influence support shall be required. Partial support, renormalization, extension, imputation, background subtraction, learned taper, and signal-derived support are excluded.
SCI-FLT-MATCHED-REQ-011	The full fixed-state response $R_t(x, y)$ and its valid domain shall remain exactly available or reconstructable; reduced kernels or beams require established invariances and an explicit measure.
SCI-FLT-MATCHED-REQ-012	“Optimal” shall mean minimum conditional variance among fixed-state linear unbiased estimators only under exact authorized inverse-covariance GLS premises; weaker weighting carries no optimality claim.
SCI-FLT-MATCHED-REQ-013	Every realization shall identify one owner-selected SCI-FLT-MATCHED-A0-001 weighting alternative separately for its parent class. No option in this draft is a default.
SCI-FLT-MATCHED-REQ-014	A parent coefficient shall retain only its declared option-specific role; name, sign, units, reciprocal, or diagonal form shall not imply covariance, precision, inverse variance, or independence.

Requirement	Normative statement
SCI-FLT-MATCHED-REQ-015	Requested, effective, observation-resolved, realized, and published state shall remain distinct. State shall be externally declared or learned once from the exact parent and external inputs, frozen, generation-identified, and applied without target-dependent update.
SCI-FLT-MATCHED-REQ-016	Observation and coadd state generations shall remain separate and shall not be substituted for one another.
SCI-FLT-MATCHED-REQ-017	Every compatible NOI member shall receive the identical frozen state, transformation, support, normalization, approximation, and failure rules as the corresponding signal realization.
SCI-FLT-MATCHED-REQ-018	NOI-informed update shall create an immutable successor generation; per-member relearning shall be a different, unavailable-in-v0.1 method.
SCI-FLT-MATCHED-REQ-019	The method shall have no automatic selector or fallback. A requested unavailable route shall fail closed without silent substitution.
SCI-FLT-MATCHED-REQ-020	When exact matching covariance authority exists, conditional covariance shall be $C_{\text{cond}} = LC_{\text{parent}}L^T$.
SCI-FLT-MATCHED-REQ-021	D^{-1} shall be called marginal conditional variance only under the exact inverse-covariance GLS premises; otherwise D is normalization.
SCI-FLT-MATCHED-REQ-022	Uncertainty availability and all U1–U7 budget terms shall be truthful and role-separated; missing covariance, cross-covariance, or entries shall be unavailable rather than zero or independent.
SCI-FLT-MATCHED-REQ-023	The public signal product shall be the role-complete atomic bundle defined in Section 8.8.
SCI-FLT-MATCHED-REQ-024	Failure of any required signal-bundle member shall fail the entire bundle; a partial bundle shall not be published as complete.
SCI-FLT-MATCHED-REQ-025	Conditional covariance, frozen-NOI uncertainty, response, and state/lineage companions shall be independently atomic and qualified by exact estimand, population, unit, support, validity, lifecycle, and named-use policy.
SCI-FLT-MATCHED-REQ-026	disabled, unavailable, failed, complete, and superseded shall remain distinct immutable lifecycle states.
SCI-FLT-MATCHED-REQ-027	No numerator, denominator, kernel, PSD, response slice, inverse scale, standardized field, or intermediate shall be public without a complete science-product role and policy.
SCI-FLT-MATCHED-REQ-028	The product contract shall preserve a first-class FLT to FRUIT handoff containing filtered quantity, template, response, valid support, units/calibration, method/state identity, uncertainty availability, and provenance without undocumented implementation state.
SCI-FLT-MATCHED-REQ-029	No FLT artifact or profile shall define FRUIT source modeling, subtraction/add-back, recurrence, learning, stopping, restart, selection, response, uncertainty, or interpretation.
SCI-FLT-MATCHED-REQ-030	Detection, candidate selection, peaks with scientific meaning, fitting, deblending, catalogs, significance, completeness, purity, morphology, and SRC ownership shall remain outside scope.

Requirement	Normative statement
SCI-FLT-MATCHED-REQ-031	Posterior/Wiener sky reconstruction, ordinary convolution, data-thresholded destriping, adaptive edge/background methods, and the historical high-pass/delta case shall remain separate methods. SCI-FLT-FIXED shall remain a separate protected package.
SCI-FLT-MATCHED-REQ-032	All six authored option families shall receive explicit scientific-owner disposition before any affected numerical, publication, covariance, handoff, persistence, or validation route is available.
SCI-FLT-MATCHED-REQ-033	Any approximate realization shall pass one selected SCI-FLT-MATCHED-A0-002 envelope against the exact N/D operator over its complete declared validity domain; iteration count or tail cap alone is not evidence.
SCI-FLT-MATCHED-REQ-034	Scientific authority, implementation conformity, validation, achieved performance, readiness, and production shall be recorded as separate claims with separate evidence and owners.
SCI-FLT-MATCHED-REQ-035	FLT shall own producer facts, named-use policies, required actions, and failure semantics; VAL may register and evaluate immutable profiles but shall not invent them.
SCI-FLT-MATCHED-REQ-036	Parent and template calibration dependence shall remain joint. Point-source flux, matched-filter beam/solid angle, extended-source fidelity, and calibration covariance require exact separate authority.
SCI-FLT-MATCHED-REQ-037	A coadd application shall retain exact contributing-observation membership and parent provenance without asserting filter/coadd commutation.
SCI-FLT-MATCHED-REQ-038	Every product and companion shall bind the full realization tuple, immutable provenance, and selected persistence/reconstruction contract needed for audit and reanalysis.
SCI-FLT-MATCHED-REQ-039	This Stage B draft and every unselected option shall make no implementation, conformity, validation, achieved-performance, readiness, production, scientific-freeze, or Unity claim.

12 Falsifiable scientific predictions

Each prediction is conditional on its cited assumptions and is falsified by one conforming counterexample within the declared validity domain. A failed prediction falsifies the affected scientific or conformity claim; it does not authorize a fallback.

Prediction	Falsifiable statement	Primary assumptions
SCI-FLT-MATCHED-PRED-001	An exact unit-amplitude matching template at admitted y gives $R_t(y, y) = 1$.	003–006
SCI-FLT-MATCHED-PRED-002	At fixed state, scaling a matching noiseless template by A scales the estimate by exactly A .	003–006
SCI-FLT-MATCHED-PRED-003	For conditionally zero-mean noise and a matching template, $E[\hat{A}(y) \mid \text{state}_g] = A$.	003–007

Prediction	Falsifiable statement	Primary assumptions
SCI-FLT-MATCHED-PRED-004	A location missing any influence-support dependency is never published as a numerical amplitude.	004,006
SCI-FLT-MATCHED-PRED-005	Nonfinite, nonreal-beyond-tolerance, zero, or negative D never produces amplitude zero as a substitute.	005–006
SCI-FLT-MATCHED-PRED-006	Observation-parent and coadd-parent applications with otherwise similar inputs retain distinct parent and state identities.	001,010
SCI-FLT-MATCHED-PRED-007	Every compatible NOI member has the same fixed L , valid support, normalization, and failure decisions as its signal realization.	010–011
SCI-FLT-MATCHED-PRED-008	Under exact matching covariance authority, empirical covariance of repeated fixed-state realizations converges to $LC_{\text{parent}}L^T$ within declared sampling uncertainty.	008–011
SCI-FLT-MATCHED-PRED-009	Without exact inverse-covariance GLS premises, removing any D^{-1} variance label changes no amplitude value but removes the unsupported uncertainty claim.	005,008–009
SCI-FLT-MATCHED-PRED-010	Any approximation accepted under SCI-FLT-MATCHED-A0-002 satisfies every normalization, unit-response, response, support/null, and covariance bound on the full test domain.	012–013
SCI-FLT-MATCHED-PRED-011	Changing a scientifically consequential realization field changes the realization or method identity and never mutates an existing complete product.	010, 013, 015
SCI-FLT-MATCHED-PRED-012	Failure of one required signal-bundle member prevents a complete signal bundle.	015
SCI-FLT-MATCHED-PRED-013	Failure of an optional covariance or NOI companion leaves a complete signal bundle complete unless its exact named-use policy requires that companion.	009, 013, 015
SCI-FLT-MATCHED-PRED-014	The output unit is always $[m]/[t]$; a rescaling $t \mapsto ct$ maps $\hat{A} \mapsto \hat{A}/c$ and preserves the modeled signal $\hat{A}t$.	003,014
SCI-FLT-MATCHED-PRED-015	A response reduced to one kernel fails its validity predicate when tested where the required invariances do not hold.	002,013
SCI-FLT-MATCHED-PRED-016	Requesting an unavailable weighting, covariance, state, approximation, handoff, or profile route never silently selects another one.	015
SCI-FLT-MATCHED-PRED-017	A filtered signal bundle plus its selected SCI-FLT-MATCHED-A0-005 lineage can answer every documented FLT-side handoff query without undocumented implementation state.	013,015
SCI-FLT-MATCHED-PRED-018	No conforming output alone can be interpreted as a detection, catalog entry, posterior sky reconstruction, FRUIT result, achieved-performance claim, readiness decision, or production decision.	001–015

13 Unselected authored option sets

All alternatives in this section are scientifically bounded candidates. They share the exact identities, assumptions, consequences, observables or bounds, failure rules, and validation consequences in both contract views because this section is imported verbatim by both. None is selected. Every parameter written as “owner-set” remains a required later disposition.

13.1 SCI-FLT-MATCHED-A0-001: Weighting and noise authority

Observation and coadd parents require separate dispositions and state generations even if the same alternative identity is selected.

Every alternative shall explicitly bind the map chart, WCS metric, sampling, units, fixed windows, complete influence support, and boundary/edge rule. It shall also state whether Fourier transformation, averaging, and radialization are used. A used operation requires its phase, normalization, measure, conjugacy or multiplicity, population, and aggregation conventions as applicable; an unused operation is recorded as absent and supplies no implied default. These declarations are part of the option identity even when the operation is not applicable to its mathematical form.

Every alternative shall also bind its estimator or external-declaration identity and version, declaring or learn-once algorithm, immutable inputs and provenance, population and conditioning state, sufficient statistics or theoretical authority, state generation, and estimator tolerances. For SCI-FLT-MATCHED-A0-001-A and SCI-FLT-MATCHED-A0-001-B, this includes the parent-covariance estimator/model from which Q_x is authorized or derived; for SCI-FLT-MATCHED-A0-001-C, it includes the spectral estimator and averaging population; for SCI-FLT-MATCHED-A0-001-D, it includes the weaker-weight estimator or exact external declaration. A generic label such as “authorized covariance” or “learned once” does not satisfy this closure.

13.1.1 SCI-FLT-MATCHED-A0-001-A: exact inverse-covariance GLS

Object and assumptions.

Q_x is the authorized inverse, or constrained Moore–Penrose inverse, of exact C_{parent} on the complete admitted support and estimable subspace. Population, conditioning, chart, sampling, units, constraints, rank, null space, and any regularization are identical for C_{parent} and Q_x . Regularization is admissible only as part of the authorized covariance model or an exact declared constraint, not as a hidden numerical convenience. The parent-covariance estimator/model provenance and the inverse-construction identity are bound as required by the family rule.

Parent coefficient.

A coefficient may enter only if covariance authority defines its exact place in C_{parent} or the inner-product measure. It is not independently reinterpreted as precision.

Consequences.

This is the sole alternative that can carry the GLS minimum-conditional-variance claim and D^{-1} marginal conditional variance, subject to every premise. Response is the full $L_x t_y$; uncertainty is $LC_{\text{parent}} L^T$. Matching-template amplitude unbiasedness still requires zero conditional mean and complete support.

Observables and bounds.

On the estimable parent subspace P_{in} , require $\|P_{\text{in}}(Q_x C_{\text{parent}} - I)P_{\text{in}}^T\| \leq \tau_{\text{inv}}$, identical declared null spaces, $e_D \leq \tau_{\text{id}}$, and unit response within τ_{id} , where the norm and owner-set numerical-identity tolerances are bound to the realization. A nonzero scientific approximation instead routes to SCI-FLT-MATCHED-A0-001-B and SCI-FLT-MATCHED-A0-002.

NOI, products, and failure.

Compatible NOI members use the same frozen Q_x . Exact covariance authority and the selected SCI-FLT-MATCHED-A0-003 representation are qualified companions, not inferred signal fields. Missing authority, population mismatch, rank/null disagreement, exceeded closure tolerance, or invalid D makes the covariance/optimality route unavailable or the attempt failed; no fallback occurs.

Validation consequence.

Validate inverse closure, null-space identity, matching-template response, exact/approximation identity, fixed-NOI parity, and covariance propagation over the full domain. A pointwise diagonal check is insufficient.

13.1.2 SCI-FLT-MATCHED-A0-001-B: structured covariance-derived weighting**Object and assumptions.**

An authorized covariance family $C(\theta, g)$ with declared population and state generation is represented by a structured $Q_x(\theta, g)$ that approximates its inverse. Structure may be block, sparse, separable, locally stationary, low-rank-plus-residual, or other declared form; the exact structure, chart, units, window, boundaries, rank, null space, and regularization are part of the option realization. Observation and coadd models are independently learned or declared. Their covariance- estimator/model provenance and structured-inverse construction are immutable option fields.

Parent coefficient.

A coefficient may be an explicitly modeled factor, mask, or measure in $C(\theta, g)$, but it gains no covariance meaning beyond that authorized model.

Consequences.

N/D preserves fixed-state matching-template unit response when valid. “Optimal” and D^{-1} variance are unavailable unless the structure is proven exact, in which case SCI-FLT-MATCHED-A0-001-A is the proper identity. Conditional covariance, if authorized, is still $LC_{\text{parent}}L^T$ and not the inverse of a displayed weight.

Observables and bounds.

Declare the complete model-fit population and residual statistic; bound inverse-action residual $\|P_{\text{in}}(Q_x C_{\text{parent}} - I)P_{\text{in}}^T\| \leq \epsilon_Q$ and every SCI-FLT-MATCHED-A0-002 metric on the declared domain. ϵ_Q , norm, sampling, aggregation, unresolved modes, and regularization path are owner-set realization fields.

NOI, products, and failure.

The frozen structured state and its generation apply identically to compatible NOI members. A structure summary may be a qualified state companion under SCI-FLT-MATCHED-A0-004; it is not a public PSD or covariance unless assigned that role. Model-population mismatch, uncovered modes, exceeded residuals, nonpositive normalization, or reconstruction failure is typed unavailability/failure with no substitution.

Validation consequence.

Validate structure-specific reconstruction, inverse-action residuals, response across nonstationary and edge domains, support/null equality, SCI-FLT-MATCHED-A0-002 bounds, uncertainty labels, and NOI parity.

13.1.3 SCI-FLT-MATCHED-A0-001-C: radially symmetrized average map-noise PSD weighting**Object and population.**

This is the explicitly examined historical candidate, not a default. Choose a declared noise-estimation population $\{n_j\}_{j=1}^J$: either externally authorized noise fields, or a declared learn-once field derived from the exact parent. The latter is weaker spectral weighting unless separate authority proves a

noise-covariance interpretation. No target, source, candidate, or NOI member may update it.

Fourier, WCS, units, and estimator.

Declare one map chart with physical coordinate r_z , WCS metric, sampling measure $\mu(z)$, transform phase $\exp(-2\pi i k \cdot r_z)$, frequency units, normalization α , and a fixed predeclared window $W(z)$. For complete supported fields define

$$F_j(k) = \alpha \sum_z W(z) n_j(z) e^{-2\pi i k \cdot r_z} \mu(z), \quad \bar{P}_b = \frac{\sum_{j,k \in B_b} w_{jk} |F_j(k)|^2}{\sum_{j,k \in B_b} w_{jk}}.$$

Radial bin B_b is defined by an owner-declared metric radius $\rho(k)$, closed bin edges, mode multiplicity, conjugate-mode convention, member weights w_{jk} , and averaging order. The units of $\bar{P}_b$ follow exactly from $[n]^2$, μ , α , and the frequency measure and shall be recorded. No background subtraction, extension, padding, adaptive taper, or signal-derived window is introduced. The spectral-estimator identity/version, immutable field provenance, sufficient statistics, and state-generation lineage are bound to this definition.

Realized weight, support, and regularization.

On admitted modes, $q(k) = (\bar{P}_{b(k)} + \lambda_b)^{-1}$ times any separately declared dimensionless mode mask, and Q_x is the declared transform-domain action restricted to complete influence support. Every λ_b , excluded mode, window dependency, boundary model, rank, and null space is fixed in g . Zero/negative/nonfinite bin estimates and bins without declared effective sample size are unavailable, not infinite precision.

Parent coefficient.

A parent coefficient may contribute only through a predeclared member weight w_{jk} , sample measure, or fixed map-domain factor whose statistical meaning and units are stated. It does not become covariance or precision merely because q is reciprocal.

Consequences.

Radialization deliberately discards angular spectral structure. It does not prove stationarity, isotropy, covariance authority, or GLS optimality. N/D remains matching-template normalized; D is not variance. Response includes the window, radial binning, null modes, boundaries, and position dependence. C_{cond} needs independent parent-covariance authority.

Quantitative observables.

Record per-bin effective sample size, dispersion before radialization, angular anisotropy statistic, member-to-member dispersion, window leakage matrix or bound, null-mode set, and regularization fraction $\lambda_b/\bar{P}_b$. Owner disposition shall set admissible bounds and the complete test population; all SCI-FLT-MATCHED-A0-002 metrics also apply.

NOI, products, and failure.

Compatible NOI members receive the same frozen $\bar{P}_b$, window, modes, support, and normalization. A PSD may be a qualified state artifact only with exact meaning, units, population, lifecycle, and policy. Missing population authority or conventions, invalid bins, failed anisotropy/leakage bounds, incomplete support, or exceeded SCI-FLT-MATCHED-A0-002 bounds makes this route unavailable/failed without fallback.

Validation consequence.

Recompute the declared sufficient statistics from the bound population, test transform normalization and units, radial-bin membership, phase and conjugacy, anisotropy and leakage bounds, support erosion, unit response, full response, uncertainty nonclaims, and fixed-NOI parity.

13.1.4 SCI-FLT-MATCHED-A0-001-D: declared weaker positive-semidefinite weighting

Object and assumptions.

Q_x is an externally declared or learn-once, frozen, positive-semidefinite linear weighting with explicit

scientific purpose but no covariance ancestry claim. Its coordinate action, units, support, boundaries, window, rank, null space, regularization, state population, and generation are complete. This alternative includes nonradial spectral or map-domain reliability weighting that is not authorized as covariance-derived. Its external declaration or learn-once estimator, version, immutable inputs, provenance, sufficient statistics, and tolerances are bound to the generation.

Parent coefficient.

A coefficient is only a declared reliability factor, mask, or part of the inner-product measure. It is not variance or precision.

Consequences.

Exact N/D gives fixed-state matching-template unit response on valid support. Minimum variance, D^{-1} uncertainty, and covariance interpretation are unavailable. Response may be nonlocal and position-dependent; uncertainty requires independent authority.

Observables and bounds.

Require Hermitian symmetry to τ_{herm} , nonnegative quadratic form to $-\tau_{\text{psd}}$, declared rank/null closure, and all selected SCI-FLT-MATCHED-A0-002 metrics. Tolerances, probe domain, and aggregation are owner-set fields.

NOI, products, and failure.

The identical frozen operator applies to compatible NOI members. The weight may be a qualified state companion under SCI-FLT-MATCHED-A0-004, not a noise product by implication. Failed positivity, undocumented coefficients, state drift, incomplete support, invalid D , or uncertainty mislabeling causes typed unavailability/failure and no fallback.

Validation consequence.

Validate linearity, Hermitian/positive- semidefinite bounds, rank/null identity, full response, support, normalization, uncertainty labels, and NOI parity across both parent classes separately.

13.2 SCI-FLT-MATCHED-A0-002: Quantitative approximation envelope

All alternatives compare one approximate realization with the exact N/D operator using Section 9.5. The test set shall cover every declared parent class, state stratum, valid-support geometry, boundary class, template phase/anchor class, rank/null regime, and response query used by a named consumer. A finite sample is acceptable only with a declared covering argument and regularity bound that extends the measured maximum to the full domain.

13.2.1 SCI-FLT-MATCHED-A0-002-A: exact-identity envelope

Assumptions and bounds.

No scientific approximation is admitted. Require $\tilde{L} = L$, $\tilde{D} = D$, $\tilde{R}_t = R_t$, $\tilde{\mathcal{V}} = \mathcal{V}$, and identical null spaces. Finite representation differences may not exceed a bound τ_{id} derived from and no larger than the selected numerical-identity tolerance for the archived realization; they may not change any classification, response, or published digit claimed as significant. If covariance is published, require $\tilde{C}_{\text{cond}} = C_{\text{cond}}$ to the same identity contract.

Test domain and aggregation.

Exhaust the finite declared domain or use a proof-producing equivalence over all omitted points and modes. Aggregation is the maximum; unresolved modes are not allowed.

Consequences.

The approximate label is only a realization detail; the scientific operator is exact under the numerical identity contract.

Failure and method boundary.

Any discrepancy above τ_{id} , support/null disagreement, or unproved mode fails this option. A change

to estimand, support, null space, state-learning rule, or failure semantics is a new method; any other numerical-identity change is at least a new realization and generation.

Validation consequence.

Archive exact-versus-realized fieldwise maxima, coverage proof, response closure, covariance closure when applicable, and classification equality.

13.2.2 SCI-FLT-MATCHED-A0-002-B: strict uniform scientific envelope

Assumptions and quantitative bounds.

On every point and declared mode of $\mathcal{T}$, require

$$\sup e_D \leq 10^{-3}, \quad \sup e_A \leq 10^{-3}, \quad \frac{\|P_{\text{out}}(\tilde{L} - L)P_{\text{in}}^T\|_2}{\|P_{\text{out}}LP_{\text{in}}^T\|_2} \leq 10^{-3}, \quad e_C \leq 2 \times 10^{-3}$$

when covariance is an authorized product. Require $\Delta_{\text{supp}} = \Delta_{\text{null}} = 0$. The response residual shall also obey $\|P_{\text{out}}(\tilde{R}_t - R_t)P_{\text{anc}}^T\|_2 / \|P_{\text{out}}R_tP_{\text{anc}}^T\|_2 \leq 10^{-3}$. Zero denominators use a dimensionally matched owner-set absolute scale for the affected operator, covariance, or response, tied to the declared input, output, and template-amplitude conventions, never a silently chosen floor.

Test domain and aggregation.

P_{in} , P_{out} , P_{anc} , their codomains, domain sampling, absolute scales, state strata, and a covering proof are fixed before evaluation. All quoted bounds are hard maxima or spectral norms; no percentile may hide an outlier. There are no unresolved output-affecting modes.

Consequences.

The realization is a bounded approximation to the same estimand. The bound is an operator-conformance statement, not evidence of astrophysical adequacy, optimality, or achieved sensitivity. The U4 budget records the full bound; covariance consumers also inherit the e_C bound.

Failure and method boundary.

One exceeded bound, changed support/null set, missing exact comparator, uncovered state stratum, or failed covering proof fails the realization. Changing a numerical technique within these bounds creates a new realization. Changing the envelope, estimand, support, null space, data-dependent selection, state learning, or failure policy creates a new method version requiring owner disposition.

Validation consequence.

Evaluate all metrics with immutable exact and approximate identities, report worst-case location/mode and signed residual, repeat for both parent classes separately, and test matching-template injection and authorized covariance propagation.

13.2.3 SCI-FLT-MATCHED-A0-002-C: named-use stratified envelope

Assumptions and quantitative bounds.

A named use declares a primary response region $\mathcal{T}_{\text{core}}$ containing at least 99% of the declared exact response energy under its explicit measure, plus the remaining valid region $\mathcal{T}_{\text{tail}}$. Require everywhere $e_D, e_A \leq 10^{-2}$ and exact support/null equality. Require response-energy RMS residual at most 10^{-2} on the core and integrated absolute response residual at most 2×10^{-2} of total exact response magnitude on the tail. If covariance is authorized, require $e_C \leq 5 \times 10^{-2}$ on each declared consumer projection. These are ceilings; the owner may select tighter profile-specific bounds but not looser ones under this identity.

Test domain and aggregation.

Core/tail measure, energy definition, consumer projections, sample covering, absolute zero-response scale, and state strata are immutable profile fields. Every valid location is classified; an unresolved mode makes the dependent named use unavailable.

Consequences.

This option permits larger, explicitly budgeted response and covariance discrepancy for a named use while preserving amplitude, support, and null semantics. It cannot support an unprofiled consumer and does not establish global optimality or performance.

Failure and method boundary.

Exceeding any ceiling, changing the core/tail partition after seeing results, concealing an unresolved mode, or using the result outside its named profile fails the route. New partition, ceilings, consumer projection, estimand, support/null, or failure semantics create a new method/profile version; a numerically changed realization under the same frozen profile is a new realization.

Validation consequence.

Report hard pointwise maxima, core energy RMS, tail integrated residual, covariance projection residuals, coverage proof, and every unavailable mode. Validate each named use independently; aggregate pass cannot erase a failed subprofile.

13.3 SCI-FLT-MATCHED-A0-003: Conditional covariance representation

Every available alternative starts from independently authorized C_{parent} for the same population, support, response, state, and units as L . It represents $C_{\text{cond}} = LC_{\text{parent}}L^T$; it does not infer covariance from D , Q , or an NOI companion. Calibration terms and cross-terms are included only with exact authority.

For each available alternative, the conditional-covariance estimator is the declared fixed-state propagation $LC_{\text{parent}}L^T$, with only the selected SCI-FLT-MATCHED-A0-002 discrepancy where that alternative expressly permits it. The companion shall bind the independently authorized parent-covariance estimator or model identity and version, declaring or learning algorithm, immutable inputs and provenance, population and conditioning, sufficient statistics or theoretical authority, state generation, and every estimator tolerance. SCI-FLT-MATCHED-A0-003-E shall instead identify which of those estimator, model, provenance, population, or conditioning authorities is missing. No representation option supplies a covariance estimator by implication.

13.3.1 SCI-FLT-MATCHED-A0-003-A: exact explicit covariance**Representation.**

Materialize every authorized covariance entry on the complete valid output domain with exact indexing, symmetry, units, population, conditioning state, rank, null space, response and generation. No correlation is omitted; authorized calibration blocks are explicit.

Approximation and consumers.

None beyond the selected numerical identity tolerance. All consumers whose domain and conditioning match may use the matrix subject to named policy. Storage is quadratic in admitted samples.

Lifecycle and failure.

The covariance is an independently atomic qualified companion. Any missing entry, broken symmetry/positive-semidefinite bound, rank/null mismatch, unit or population mismatch, or incomplete atomic member makes it failed or unavailable; it never fills unknown entries with zero.

Validation consequence.

Recompute selected entries and quadratic forms from L and C_{parent} , exhaust index/symmetry closure, verify units, rank/null space, calibration blocks, and match empirical fixed-state ensembles within declared sampling uncertainty.

13.3.2 SCI-FLT-MATCHED-A0-003-B: exact structured covariance

Representation.

Persist an exact factorization or structured operator whose queries reproduce every authorized entry and quadratic form of $LC_{\text{parent}}L^T$ on the full valid domain. Structure, factors, index maps, support, response, rank/null, regularization identity, population, units, calibration blocks, and state generation are immutable.

Approximation and consumers.

Scientific approximation is zero; a reconstruction tolerance τ_C is an owner-set numerical-identity field. All declared entry, marginal, block, draw, and quadratic-form queries are supported; any unimplemented query is explicitly unavailable rather than approximated. Storage is structure-dependent and less than full materialization only when the exact structure permits it.

Lifecycle and failure.

The structured state is an atomic companion. Failed reconstruction, query-dependent drift, unbound environment facts, rank/null mismatch, omitted calibration term, or residual above τ_C fails the companion and dependent consumers.

Validation consequence.

Use adversarial and basis-spanning queries to compare with direct $LC_{\text{parent}}L^T$, verify deterministic reconstruction and all metadata, and test every advertised consumer query.

13.3.3 SCI-FLT-MATCHED-A0-003-C: authorized projected covariance

Representation.

For one named consumer projection P_c , publish the complete exact or SCI-FLT-MATCHED-A0-002-bounded covariance $P_cLC_{\text{parent}}L^TP_c^T$. The projection basis, measure, domain, support, response, retained rank/null space, state, units, population, calibration treatment, and reconstruction bound are immutable. All correlations outside the projection are explicitly unknown.

Approximation and consumers.

The owner selects exact projection or an owner-set residual ceiling $\epsilon_{C,c}$ not exceeding the selected SCI-FLT-MATCHED-A0-002 covariance limit. Only the named projected consumer is admissible. No diagonal, map-wide, independence, draw, or inverse-covariance use follows.

Lifecycle and failure.

Each projection is an independently atomic companion. Missing basis vectors, changed projection, rank loss, unbounded omitted modes, calibration mismatch, or exceeded residual makes that consumer route unavailable/failed without changing the signal bundle.

Validation consequence.

Validate projection orthogonality/measure, rank/null identity, exact projected entries and quadratic forms, residual bounds, omitted-mode labels, and rejection of every unlisted consumer.

13.3.4 SCI-FLT-MATCHED-A0-003-D: exact lineage-resolvable covariance on demand

Representation.

Persist no covariance values, but retain exact immutable lineage sufficient to reconstruct L , C_{parent} , domain/support, response, units, population, state, rank/null space, regularization, calibration terms, and the selected computation identity. Reconstruction must not require undocumented implementation state.

Approximation and consumers.

Each supported query is reconstructed exactly within owner-set τ_C and yields an immutable qualified result with its own lifecycle. Supported query classes and resource limits are declared; unsupported queries are unavailable. This is not permission to substitute a different covariance model later.

Lifecycle and failure.

The lineage closure is the atomic companion until a query result is produced. Missing inputs, nonreproducible environment facts, version drift, failed reconstruction test, or residual above τ_C makes the covariance route unavailable/failed.

Validation consequence.

Before publication, perform and archive a representative basis-spanning reconstruction, hash/identity closure, repeated environment-independent result, and comparisons with direct propagation for all advertised query classes.

13.3.5 SCI-FLT-MATCHED-A0-003-E: covariance unavailable**Representation and assumptions.**

No authorized matching C_{parent} or defensible representation exists. Record the exact missing authority or premise. No numerical covariance, marginal variance, standardization, significance, or independence statement is published.

Consequences and consumers.

The filtered amplitude signal may remain complete if all signal requirements pass. Every consumer requiring C_{cond} is unavailable. A separately qualified frozen-NOI companion does not replace it.

Lifecycle and failure.

This is typed unavailability, not a failed attempt and not an all-zero matrix. Later authority creates a successor companion or generation without mutating the predecessor.

Validation consequence.

Validate absence of covariance-valued fields, D^{-1} variance labels, zero-filled matrices, significance products, and consumers that silently assume independence.

13.4 SCI-FLT-MATCHED-A0-004: Immutable state persistence

Every alternative binds the state-generation identifier, learning/declaring algorithm and version, exact parent and external inputs, learning population, template, Q_x , windows, rank/null, regularization, support, numerical identity, environment-dependent facts, failure policy, and compatible-NOI parity. Updates create successors; predecessors are not rewritten.

13.4.1 SCI-FLT-MATCHED-A0-004-A: full state materialization**Persistence.**

Persist every realized state element used by L , response, support, normalization, uncertainty, and named-use policy, together with exact indexing, units, hashes, algorithm/version, inputs, population, and generation. No reconstruction from learning data is needed. Storage scales with full state.

Numerical identity and reconstruction test.

Reloaded state shall match the archived materialization bit-for-bit or through a lossless canonical serialization; application results shall match within owner-set τ_{state} no larger than the chosen SCI-FLT-MATCHED-A0-002 identity bound.

Audit, reanalysis, and NOI.

This gives the broadest audit and reanalysis independence from the original environment. Every compatible NOI member binds the same archived state hash. A successor has a new hash and generation.

Failure.

Missing element, failed hash, ambiguous indexing/unit, environment override, or reconstruction residual above τ_{state} fails state admission and all dependent products; no relearning fallback.

Validation consequence.

Round-trip every field, compare operator and response on a basis-spanning domain, verify NOI state hashes, and demonstrate predecessor immutability after successor creation.

13.4.2 SCI-FLT-MATCHED-A0-004-B: structured compact state**Persistence.**

Persist a declared compact factorization, parameterization, or sufficient-state representation plus exact reconstruction semantics. Bind all factors, index maps, conventions, tolerances, algorithm/version, original inputs/population, environment facts, and generation. Storage is structure-dependent; discarded information is forbidden unless proved irrelevant to every supported query and operator action.

Numerical identity and reconstruction test.

Reconstruct full operator actions, response queries, support/null decisions, and normalizations within owner-set τ_{state} and the selected SCI-FLT-MATCHED-A0-002 envelope on a complete covering domain. Unsupported queries are explicitly unavailable.

Audit, reanalysis, and NOI.

Audit/reanalysis is limited to the advertised query set. Compatible NOI members bind the identical compact-state identity. Changing structure, factors, tolerance, or query set creates a successor generation and may create a new method under SCI-FLT-MATCHED-A0-002.

Failure.

Undeclared discarded state, nonunique reconstruction, unbound environment dependence, uncovered query, state drift, or exceeded residual fails the route without full-state or relearning fallback.

Validation consequence.

Use basis-spanning and adversarial queries, support/null boundary cases, repeated reconstruction across declared environments, hash closure, and NOI parity. Report unsupported queries.

13.4.3 SCI-FLT-MATCHED-A0-004-C: exact lineage reconstruction**Persistence.**

Persist the complete immutable parent and external-input identities, learning population, template, algorithm/version, declared environment, numerical libraries or mathematical identities as scientifically consequential, random seeds when any, tolerances, failure policy, and lineage needed to rerun *DeclareOrLearnOnce*. No undocumented state is allowed.

Numerical identity and reconstruction test.

Before any dependent product is complete, reconstruct the state independently and require canonical state, operator, response, support/null, and normalization agreement within owner-set τ_{state} and the selected SCI-FLT-MATCHED-A0-002 envelope. Bitwise identity is required for fields declared discrete or exact.

Audit, reanalysis, and NOI.

Reanalysis depends on continuing availability of every lineage input and authorized environment. NOI members reference the reconstructed frozen state identity, never rerun learning per member. Any NOI-informed update creates a successor lineage.

Failure.

Missing/mutable input, unavailable algorithm version, unreproducible environment, stochastic ambiguity, failed closure, or exceeded tolerance makes state and dependent routes unavailable/failed; no inferred substitute or relearning fallback exists.

Validation consequence.

Perform two independent reconstructions, close all identities/hashes, compare complete operator/response cover, test archival availability and environment declarations, and demonstrate identical

NOI use.

13.5 SCI-FLT-MATCHED-A0-005: Response, product, and FLT to FRUIT interface

Every alternative retains the same role-complete signal bundle and preserves the exact operator, filtered quantity, template, full response meaning, valid support, units/calibration, method/state identity, uncertainty availability, and provenance. The choice concerns response/state representation and query availability, not the scientific response itself and not FRUIT behavior.

13.5.1 SCI-FLT-MATCHED-A0-005-A: full response materialization

Representation and storage.

Materialize $R_t(x, y)$ for every admitted output x and every declared template anchor y , with exact index maps, measure, units, valid support, state/operator identity, template generation, and any authorized calibration relation. Storage is generally quadratic in the two response domains; no invariance reduction is assumed.

Exactness and queries.

Values match direct $L_x t_y$ within owner-set τ_R no larger than the selected SCI-FLT-MATCHED-A0-002 response bound. Entry, slice, integral, moment, beam/solid-angle, or other queries are supported only when their measure and validity policy are declared. A materialized matrix does not itself authorize a reduced scientific interpretation.

Consumers and handoff.

Any named consumer, including a future FRUIT contract, may use only its registered query/profile. The signal bundle and response companion supply all FLT-side identity without undocumented state.

Lifecycle and failure.

Response is an independently atomic companion or required handoff member according to named policy. Missing entries, index/unit mismatch, invalid-support leakage, exceeded τ_R , or incomplete lineage fails response-dependent routes; it does not define FRUIT behavior or retroactively change the signal unless required by that signal policy.

Validation consequence.

Exhaust indexing and support, compare basis slices with direct $L_x t_y$, test asymmetry/nonstationarity retention, units, calibration references, query policies, and atomic lifecycle.

13.5.2 SCI-FLT-MATCHED-A0-005-B: exact structured response with documented query contract

Representation and storage.

Persist a compact exact factorization or mathematical response representation plus state needed to answer a finite declared query vocabulary. Bind operator/template/state versions, index maps, coordinate measures, support/null rules, units/calibration, approximation identity, and provenance. Storage follows the structure; no stationarity, isotropy, symmetry, or locality is presumed without proof.

Exactness and queries.

Every advertised query reconstructs $L_x t_y$ within owner-set τ_R and the selected SCI-FLT-MATCHED-A0-002 envelope. Query vocabulary may include entries, local slices, named projections, or response-applied vectors. Unadvertised queries and invalid locations are typed unavailable.

Consumers and handoff.

Each consumer binds the exact query vocabulary and validity domain. A future FRUIT contract can consume the interface without implementation knowledge, but must separately define its science.

Lifecycle and failure.

The structured response and query contract are an atomic companion. Hidden environment state, unbound structure assumptions, unsupported queries presented as zeros, failed reconstruction, or response residual above τ_R fails dependent routes with no switch to a reduced kernel.

Validation consequence.

Use basis-spanning and adversarial response queries across boundaries and state strata; disprove undeclared invariances, verify exact unavailable responses for unsupported queries, and close all lineage and units.

13.5.3 SCI-FLT-MATCHED-A0-005-C: exact lineage-resolvable response on demand**Representation and storage.**

Materialize no response array. Persist the complete immutable lineage needed to reconstruct L , t_y , valid domains, support/null decisions, coordinate measure, units/calibration, state, approximation, and failure policy. This may share an SCI-FLT-MATCHED-A0-004 lineage only when that selected alternative exposes every required response fact.

Exactness and queries.

Before product completion, independently reconstruct a declared basis-spanning sample within owner-set τ_R and the selected SCI-FLT-MATCHED-A0-002 envelope. Supported on-demand query classes and resource limits are fixed. A later query creates an immutable qualified response result; unsupported or unreconstructable queries are unavailable.

Consumers and handoff.

The FLT to FRUIT interface contains the lineage identity and documented scientific reconstruction contract, never a pointer to undocumented implementation state. A future FRUIT named use must require only supported queries or commission a successor FLT product contract.

Lifecycle and failure.

Missing lineage, unavailable template/state, version/environment drift, failed basis test, exceeded τ_R , or query nondeterminism makes response-dependent handoff unavailable/failed. The filtered amplitude map may remain complete under its own policy.

Validation consequence.

Perform two independent response reconstructions, compare direct $L_x t_y$ over the covering set, verify all identities and archive availability, test query rejection, and demonstrate that no undocumented state is needed.

13.6 SCI-FLT-MATCHED-A0-006: VAL profile granularity

All alternatives preserve six scientifically distinct verdict roles: parent admission (PA), state admission (SA), public signal publication (SP), conditional covariance use (CU), optional NOI companion use (NU), and FLT to FRUIT handoff (FH). FLT supplies immutable producer facts, named-use policy, consumer action, and failure semantics. VAL only registers and evaluates the versioned profiles. An aggregate never changes a role verdict.

13.6.1 SCI-FLT-MATCHED-A0-006-A: six role-separated profiles**Profile set and facts.**

Maintain one immutable versioned FLT-owned profile for each of PA, SA, SP, CU, NU, and FH. Each contains only the producer facts, option identities, validity predicates, lifecycle states, and evidence references needed for that role.

Policy and action.

Every profile names admissible consumers and exact pass, unavailable, and failed actions. PA/SA failure prevents dependent SP; CU, NU, and FH may be unavailable without invalidating SP unless SP's named policy explicitly requires one. No role is inferred from another.

Aggregation and concealment.

A release view may display six verdicts but has no scalar pass. Combining roles would conceal scientifically distinct failures and is forbidden under this option.

Versioning and failure.

Any fact, threshold, selected AO, policy, or action change versions only the affected profile plus declared dependents. Missing facts yield typed unavailability; evaluator failure yields failed.

Validation consequence.

Test all 5^6 lifecycle combinations by rule or exhaustive generated truth table, dependency propagation, consumer action, version isolation, and proof that no CU/NU/FH failure silently changes SP.

13.6.2 SCI-FLT-MATCHED-A0-006-B: three layered profiles with mandatory subverdicts**Profile set and facts.**

Use an Admission profile containing separate PA and SA subverdicts, a Publication profile containing SP plus separate CU and NU subverdicts, and a Handoff profile containing FH. Every subverdict retains its own producer facts, named-use policy, action, lifecycle, and selected options.

Policy and action.

Layer-level action is a deterministic vector-valued rule over subverdicts. Admission blocks SP only according to explicit PA/SA dependencies. CU and NU actions affect only named consumers; FH is independent of FRUIT science and cannot upgrade SP.

Aggregation and concealment.

A displayed layer scalar may be a warning summary but cannot replace, overwrite, or be consumed without the complete subverdict vector. Combining CU or NU with SP would conceal optional-companion failure if only the scalar were retained, so scalar-only persistence is forbidden.

Versioning and failure.

Profiles and subpolicies are independently versioned; an aggregate version binds the exact component versions. Missing subverdicts make the layer unavailable; evaluation faults fail only affected subverdicts and declared dependents.

Validation consequence.

Exhaust subverdict transition tables, verify vector preservation through registry and consumer interfaces, test component version changes, and demonstrate that every distinct role failure remains visible and actionable.

13.6.3 SCI-FLT-MATCHED-A0-006-C: one composite profile with a six-role verdict vector**Profile set and facts.**

Use one immutable composite profile only if it contains six separately addressable PA, SA, SP, CU, NU, and FH verdict records, each with complete producer facts, named-use policy, action, lifecycle, evidence, and selected-option identity.

Policy and action.

Consumers request a named vector component, never a global Boolean. The composite defines an explicit dependency graph and vector-valued transition function. Unavailable and failed remain distinct per role.

Aggregation and concealment.

A scalar aggregate is scientifically inadmissible because it necessarily conceals at least one distinct role

failure. A human summary may list counts only alongside the full vector and cannot drive action.

Versioning and failure.

The composite version binds six component versions and dependency-graph version. Missing one component makes only that role and declared dependents unavailable, while corruption of the composite identity fails the entire profile record.

Validation consequence.

Prove lossless round-trip of all six verdicts, exhaust the dependency graph and action table, reject scalar-only consumers, test independent failures and version changes, and confirm VAL adds no policy.

13.7 Option-set closure and unavailable routes

Until the scientific owner selects and fully parameterizes an alternative in each family, the following routes remain unavailable: weighting realization and any corresponding optimality label (SCI-FLT-MATCHED-AO-001); approximate-operator conformity (SCI-FLT-MATCHED-AO-002); conditional-covariance publication or use (SCI-FLT-MATCHED-AO-003); state audit/reconstruction and dependent completion (SCI-FLT-MATCHED-AO-004); response-dependent named use and FLT to FRUIT handoff (SCI-FLT-MATCHED-AO-005); and registered validation/publication policy (SCI-FLT-MATCHED-AO-006). Selection of an option does not by itself establish implementation conformity, validation, achieved performance, readiness, or production.

14 Edge cases, exclusions, and failure routing

Condition	Scientific meaning	Required handling
Empty or incomplete influence support	No defined reference estimand at the affected location	Mark unavailable before evaluation; no padding, extension, imputation, partial estimate, or renormalization.
Numerical fill near an edge	Merely a computational placeholder	Admit an output only after exact erosion proves zero dependency on every filled value.
$D = 0$, $D < 0$, non-finite D , or materially complex D	Template is null, weighting is invalid, or numerical identity failed	Typed unavailability or attempt failure; never publish zero amplitude, infinite variance, or fallback.
Template lies partly in $\mathcal{N}(Q_x)$	Some declared template modes are unestimable	Allowed only if the selected option explicitly defines the same estimable subspace and $D > 0$; record response loss and null space. Otherwise unavailable.
Indefinite or non-Hermitian Q_x	Weight is outside declared quadratic-form semantics	Fail selected SCI-FLT-MATCHED-AO-001 closure unless an exact distinct operator and inner product were separately authored; no silent symmetrization.

Condition	Scientific meaning	Required handling
Zero, negative, or non-finite PSD bin	No finite declared reciprocal weighting for that mode/bin	Apply only owner-declared regularization within the frozen state; otherwise exclude as a declared null mode or make route unavailable. Never infer infinite precision.
Sparse radial bin or large angular scatter	Radially averaged PSD lacks declared information or radial sufficiency	Fail owner-set effective-sample, anisotropy, or leakage bounds; no neighboring-bin interpolation unless separately authored.
Unknown covariance entry	Scientific knowledge is absent	Preserve unknown; do not encode zero, diagonal independence, or a usable inverse.
Available Q but unavailable C_{parent}	Weighting exists without covariance authority	Amplitude filtering may proceed under a selected weaker option; optimality, D^{-1} variance, and C_{cond} are unavailable.
Available C_{parent} with mismatched population/state	Covariance does not describe this realization	Covariance route unavailable; no rescaling or population substitution.
Template rescaled by $c \neq 0$	Amplitude convention changes	Create a new template/realization identity; $\hat{A}$ rescales by $1/c$ and units/lineage must follow.
Template phase or anchor changed	Scientific response convention changes	New template and realization; never treat as a numeric implementation detail.
Template mismatch to a real signal	Base matching-template estimand does not equal an unconstrained source property	Report only response-weighted shape amplitude; no bias correction, morphology, or new uncertainty without separate authority.
Parent beam or calibration exists	Provenance is present, not inherited matched-filter meaning	Preserve lineage; require exact authority before flux, matched beam/solid angle, extended-source, or calibration-covariance claim.
Coadd membership changes	Parent identity changes	New coadd parent, state, realization, response, and products; do not reuse observation/coadd state by implication.
NOI member suggests a new state	Per-member relearning would change the ensemble method	Apply the frozen state or mark member incompatible; a learned update creates a successor generation outside the current ensemble.

Condition	Scientific meaning	Required handling
Approximation misses one null or support mode	Estimand/validity differs even if aggregate error is small	Fail every SCI-FLT-MATCHED-AO-002 alternative; no percentile or mean can hide the disagreement.
Iteration/tail cap reached	Work limit, not convergence evidence	Accept only if every selected quantitative envelope bound independently passes; otherwise fail.
Optional companion fails after signal completion	Companion atomicity failed, not necessarily signal science	Mark companion failed ; preserve signal complete unless exact named-use policy made the companion required.
Required signal member fails	Role-complete signal does not exist	Fail the entire signal bundle; do not publish a partial bundle as unavailable pixels inside an otherwise complete product.
Successor state or option is approved	Scientific identity advances	Create immutable successor generation and mark predecessor superseded only by policy; never mutate predecessor facts.
Unselected or unavailable option requested	No authorized route exists	Fail closed and name the missing owner disposition; no automatic selector or fallback.
Response query is unsupported	Scientific interface does not define that quantity	Return typed unavailability; never approximate with a kernel, radial profile, parent beam, or zero.
FLT to FRUIT consumer asks for FRUIT behavior	Request crosses package authority	Withhold the behavior and require a separate FRUIT contract; the FLT signal and handoff facts remain unchanged.
Filtered-map peak is examined	A numerical feature is not an authorized source object	No detection, candidate, significance, completeness, purity, fit, morphology, catalog, or SRC claim follows.

14.1 Explicit method exclusions

Posterior/Wiener sky reconstruction, ordinary convolution, source or candidate learning, data-thresholded destriping, automatic selectors/fallbacks, adaptive edge/background behavior, the historical high-pass/delta case, partial-support estimation, FLT-owned coaddition, source analysis, and FRUIT algorithms are not degenerate settings of this contract. They require separate scientific contracts if later commissioned.

SCI-FLT-FIXED remains a separate protected package. This contract neither selects, modifies, supersedes, nor supplies scientific authority for that method.

The historical compatibility term “Wiener filter” may refer to this method only when accompanied by the exact **SCI-FLT-MATCHED** identity and may not imply a sky prior, posterior estimand, or Wiener

reconstruction.

14.2 Nonclaim closure

Passing a scientific prediction or an engineering conformance test establishes only that named claim for the exact immutable realization. Validation-profile pass is not achieved performance. Achieved performance is not readiness. Readiness is not production. None of those layers can select an authored option, repair missing scientific authority, or broaden this validity domain.

Part III

Owner-disposition guide

15 What remains for the scientific owner

This draft develops, but does not select, six decision families. The owner must select and fully parameterize them before their dependent routes become available; the decision ledger states all 17 exact owner questions.

Family	Decision	Route unavailable before disposition
SCI-FLT-MATCHED-AO-001	Weighting/noise authority for observation and coadd separately; all population, coefficient, PSD/model, rank/null, regularization, and numeric fields	A weighting realization, any optimality label, and any variance meaning derived from the weight.
SCI-FLT-MATCHED-AO-002	Exact, strict-uniform, or named-use approximation envelope and all test-domain identity fields	Any approximate-operator conformance claim.
SCI-FLT-MATCHED-AO-003	Explicit, structured, projected, on-demand, or unavailable conditional covariance and its consumers	Conditional-covariance publication, standardization, or covariance-dependent use.
SCI-FLT-MATCHED-AO-004	Full, compact, or lineage state persistence and reconstruction tolerance	State audit/reanalysis and completion of products whose named use requires reconstructable state.
SCI-FLT-MATCHED-AO-005	Full, structured-query, or lineage response representation and query vocabulary	Response-dependent use and FLT to FRUIT handoff.
SCI-FLT-MATCHED-AO-006	Six-profile, three-layer, or composite-vector VAL granularity	Registered producer-policy evaluation and publication/handoff profile claims.

No option choice alone authorizes a numerical route. After owner disposition, an implementation still needs separate conformity evidence; profiles need separate validation; measured performance, readiness, and production remain separate decisions.

A Author-packet basis and firewall statement

This draft used only these eight admitted objects:

- `AUTHOR_PACKET_MANIFEST.md` and `SCOPE_BRIEF.md`;
- `AUTHOR_SUPERSESSION_COVER.md` and `AUTHOR_CONVENTIONS_AND_OWNERSHIP.md`;
- `SCIENTIFIC_OWNER_DECISIONS_FOR_AUTHORSHIP.md`;
- `AUTHOR_OPERATOR_STATE_AND_PRODUCT_TAXONOMY.md`; and
- `AUTHOR_BOUNDARIES.md` and `REQUIRED_AUTHORED_OPTION_SETS.md`.

Links in those objects were not opened. No implementation, configuration, schema, test, audit, validation, reduction, generated product, repository history, neighboring package, external source, or model-memory source supplied scientific content.